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By Amit Katiyar (MCA-JNU)



GENERAL ENGLISH

Directions (Q. Nos. 1 to 5) Read the following passage and answer the questions.

Anthropologists have pieced together the little they know about the history of left handedness and right handedness from indirect evidence. Though early men and women did not leave written records, they did leave tools, bones, and pictures. Stone Age hand axes and hatchets were made from stones that were carefully chipped away to form sharp cutting edges. In some the pattern of chipping shows that these tools and weapons were made by right handed people designed to fit comfortably into a right hand. Other Stone Age implements were made by or for left handers prehistoric pictures painted on the walls of caves provide further clues to the handedness of ancient people.

A right hander finds it easier to draw faces of people and animals facing toward the left whereas, a left hander finds it easier to draw faces facing toward the right. Both kinds of faces have been found in ancient painting.

On the whole the evidence seems to indicate the prehistoric people were either ambidextrous or about equally likely to be left or right handed. But in the Bronze Age, the picture changed. the tools and weapons found from that period are mostly made for right handed use. The predominance of right handedness among humans today had apparently already been established.

- What is the indirect evidence through which the preferred handedness of the Stone Age people could be understood?
 - Petrified forms of vegetation
 - Patterns of stone chipping
 - Fossilised waste material
 - Fossilised footprints
- According to the passage, a person who is right handed is more likely to draw people and animals that are facing
 - upward
 - downward
 - toward the right
 - toward the left
- What does the word 'the picture' refer to which of the following?
 - Faces of animals and people
 - people's view from inside a cave
 - People's tendency to work with either hand
 - the kinds of paints used on cave walls
- The author implies that which of the following development occurred around the time of the Bronze age.
 - The establishment of written records
 - A change in the style of cave painting
 - An increase in human skill in the handling of tools
 - The prevalence of right- handedness

- What is the main idea conveyed through the passage?
 - The purpose of ancient implements
 - The significance of prehistoric cave paintings
 - The development of right handedness and left handedness
 - The pattern of clipping ancient tools
- Which of the following refers to the idiom 'under the Sun'?
 - Anything and everything
 - A large number of things
 - A few things
 - Something
- Choose a phrasal verb to replace the explanation in brackets: When we arrive at the station, we (descend from)..... the train.
 - Get down
 - Stand down
 - Get off
 - Stand out
- Choose the suitable word from the following and fill in the blank:
The medal was awarded for the student'sconduct and courage.
 - non-receptive
 - exemplary
 - unreliable
 - disputable
- Which of the following is a correctly spelt word?
 - Hiderence
 - Hindrence
 - Hindarrence
 - Hindrance
- Which one of the following statements is grammatically correct?
 - The Earth revolves round the Sun
 - I have not seen him since four years
 - She met an one-eyed man
 - One of the books borrowed by the Students are famous
- Choose the set words from among the alternatives given. Which when inserted in the sentence best suit the meaning of the sentence?
The..... of evidence was on the side of plaintiff since all but one witness testify his story was...
 - paucity, accurate
 - prosperity, far fetched
 - preponderance, correct
 - accuracy, insufficient
- Choose the one which is nearest in meaning to the word 'TURN UP'?
 - Show up
 - Come up
 - Land up
 - Crop up
- The phrase 'ready to believe' means
 - Credulous
 - Creditable
 - Credible
 - Incredible
- Choose the appropriate word from among the choice to fill in the blank in sentence; "If you drink to much ,it will.....your judgment."
 - impair
 - impede
 - impose
 - impel
- Choose the set of words for each blank that best fits the meaning of the following sentence as a whole. green and black tea are obtained from the same plant, there are quite a few significant difference..... them.
 - Since, among
 - However, in
 - Though, between
 - Because, across





16. Choose the correct alt which substituted for the given word/sentence. can be A person who travels to a sacred place as an act of religious devotion.
(a) Hermit (b) Pilgrim
(c) Saint (d) Medicant
17. Pick out the most effective word from the given words to fill in the blank to make the sentence meaningfully complete: Some people themselves into believing that they are indispensable to the organisation they work for.
(a) keep (b) fool (c) delude (d) denigrate
18. Fill in the blanks with appropriate phrase to make the sentence meaningfully complete.
.....bad weather, the trip will be postponed to next week.
(a) In case (b) In case of
(c) In case to (d) In case from
19. In the following sentence, choose the most suitable one word for the expression: "A book containing summarised information on all branches of knowledge".
(a) Dictionary (b) Anthology
(c) Encyclopedia (d) Directory
20. Pick out the most effective word from the given words to fill in the blanks to make the sentence meaningful completely:
The man was about to move his bike into the compound of his apartment when a passes by.....down the motor cycle.
(a) forced (b) fell
(c) turned (d) knocked

COMPUTER AWARENESS

21. Which one of the following boolean algebraic rule is correct?
(a) $A \cdot A' = 1$ (b) $A + AB = A + B$
(c) $A + A'B = A + B$ (d) $A(A + B) = B$
22. The representation of a floating point binary number +1001.11 in 8-bit fraction and 6-bit exponent format is
(a) fraction : 01001110 exponent: 000100
(b) fraction : 00001001 exponent: 000011
(c) fraction : 10010000 exponent: 110000
(d) fraction : 00100100 exponent: 011000
23. Which term is redundant in the expression $AB + A'C + BC$?
(a) $B'C$ (b) $A'C$ (c) AB (d) None
24. Let the memory access time is 10 milliseconds and cache access time is 10 microsecond. Assume the cache hit ratio 15%. The effective memory access time is
(a) 2 milliseconds (b) 18.51 milliseconds
(c) 1.85 microsecond (d) 1.85 milliseconds

25. Which of the following is the representation of decimal number (-147) in 2's complement notation on a 12-bit machine?
(a) 111101101100 (b) 110001001101
(c) 111101101101 (d) 000001101101
26. The first instruction of bootstrap loader program of an operating system is stored in
(a) RAM (b) Hard disk
(c) BIOS (d) None of these
27. Consider the equation $(123)_5 = (x8)_y$ with x and y as unknown. The number of possible solutions is
(a) 1 (b) 2 (c) 3 (d) 4
28. The smallest integer that can be represented by an 8 bit number in 2's complement form is
(a) -256 (b) -128 (c) -127 (d) -255
29. Which of the following in a functionally complete set of gates?
I. NAND
II. NOR
(a) I but not II (b) II but not I
(c) Neither I nor II (d) Both I and II
30. The total number of binary function that can be defined using n boolean variables is
(a) 2^{n-1} (b) 2^n (c) 2^{n+1} (d) None

ANALYTICAL ABILITY AND LOGICAL REASONING

31. Two person S and M have made the following statement among themselves.
• S says that I am certainly not over 40 year.
• M says that I am 38 years and you are at least 5 year old than me.
• S says you are at least 39 years .
• If all the above statements are wrong, what are the ages of M and S?
(a) 36 and 40 (b) 36 and 41
(c) 37 and 40 (d) Cannot be determined
32. What is the largest number of positive integers to be picked up randomly so that the sum of difference of any two of the chosen numbers is divisible by 10?
(a) 2 (b) 5 (c) 7 (d) 10
33. Five children were administered psychological tests to know their intellectual levels. In the report psychologists pointed that child A is less intelligent than child B. The child C is less intelligent than child D. The child B is less intelligent than child C and child A is less intelligent than child E. Which child is most intelligent?
(a) D only (b) E only
(c) D or E (d) Neither D nor E
34. From a group of 7 men and 6 women, a committee of 5 persons with more males than females is to be formed. In how many ways can this be done?
(a) 564 (b) 645 (c) 735 (d) 756



35. A, B, C, D, E and F are 6 friends from a club. There are two houses wives, one lecturer, one architect, one accountant and one lawyer in the group. There are three married couples. The lawyer B is married to D, who is a house wife. No lady is either an architect or an accountant. C, the accountant is married to F, who is lecturer. If E is not a house wife, what is the profession of E?

- (a) Lawyer (b) Architect
(c) Lecturer (d) Accountant

36. There are five books A, B, C, D and E placed on a table. If A is placed below E, C is placed above D, B is placed below A and D is placed above E, then which of the following books touches the surface of the table?

- (a) C (b) B (c) A (d) E

37. The following series is obtained by considering representations of decimal 99 in different number system. The next two numbers in the sequence are 99, 90, 83, 78

- (a) 71, 69 (b) 69, 57
(c) 67, 59 (d) 69, 63

Directions (Q. Nos. 38 to 40) Questions are based on the following

- In a family of six person A, B, C, D, E and F, there are two married couples.
- D is the grandmother of A and mother of B
- C is wife of B and mother of F
- F is granddaughter of E

38. Who is C to A?

- (a) Daughter (b) Mother
(c) Father (d) Cannot be determined

39. Which of the following is true?

- (a) A is brother of F (b) A is sister of F
(c) B has two daughters (d) None of these

40. What among the following is one of the couples?

- (a) CD (b) DE (c) EB (d) None

41. The missing number of the following series 336, 210, 120, 60, , 6 is

- (a) 24 (b) 30 (c) 34 (d) 40

42. If the day after the day after tomorrow is three days Before Friday, then today is

- (a) Tuesday (b) Thursday
(c) Saturday (d) Monday

43. Find the missing term of the following series DCXW, HGTS,, POLK, TSHG

- (a) KLOP (b) LKOP
(c) KLPO (d) LKPO

44. Four passengers in a train find that they are from an interesting group. Two of them are lawyers and the other two are doctors. Two of them speak Bengali and the other two speak Hindi and no two of the same profession speak the same language. They also found that two of them are Christians and two are Muslims and no two of the same religion speak the same languages. The Hindi speaking doctor is a Christian. Then, which of the following statement logically follows?

- (a) The Bengali speaking lawyer is a Muslim
(b) The Christian lawyer speaks Bengali
(c) The Bengali speaking doctor is a Christian
(d) The Bengali speaking doctor is a Muslim

Directions (Q.Nos. 45 to 47) Questions are based on the following.

In an amusement park seven friends-Feroz, Gautam Harish, Javed, Kumar, Laxman and Mohan are deciding who will ride the roller coaster. There is time for only one ride before the park closes.

- If Feroz rides Gautam must ride.
- If Gautam and Harish both ride, Javed cannot ride.
- If Harish and Javed both ride, Laxman cannot ride.
- If Javed rides, either Kumar or Mohan must ride.
- Kumar and Laxman cannot both ride, but one of them must ride.
- Kumar and Mohan cannot both ride.

45. Which of the following is an acceptable combination of riders if only three people ride?

- (a) Harish, Javed and Laxman
(b) Harish, Javed and Kumar
(c) Feroz, Gautam and Javed
(d) Gautam, Kumar and Laxman

46. If Javed and Mohan both ride, which of the following is true?

- (a) Gautam cannot ride (b) Harish must ride
(c) Feroz cannot ride (d) Laxman must ride

47. If Feroz and Harish both ride, what is the greatest number of people who can ride?

- (a) 5 (b) 7 (c) 4 (d) 6

48. The numbers is squares in the following 4×6 grid is

- (a) 36 (b) 44 (c) 51 (d) 54





49. A cube is made up of 125, One cm, smaller cube placed on a table. How many smaller cubs are visible only on three sides?
(a)4 (b)8 (c)12 (d)16
50. Three thieves rob a bakery of bread, one after the other. Each thief takes half of what is present and half bread. If 3 breads remain at the end, what is the number of breads that were present initially?
(a) 24 (b) 31 (c) 37 (d) 41
51. A caterpillar crawls up a pole of 75 inches high standing from the ground. Each day it crawls up 5 inches and each night it slides down 4 inches. When will it reach the top of the pole?
(a)At the end of 70 days (b)At the end of 71 days
(c)At the end of 72 days (d)At the end of 73 days
52. A man's investments doubles in every 5 years. If the invested RS/- 5,000 in each of the years 1990, 1995, 2000 and 2005, then what was the total amount received by him in 2010?
(a)RS/- 140000(b)RS/- 30000
(c)RS/- 70000(d)RS/- 150000

Directions (Q. Nos.53 to 57) Questions are based on the Following information.

A, B, C, D, E, F, G and H are sitting around a circular table facing the centre. Each one of them has a different profession viz, doctor, engineer, architect, teacher, clerk, shopkeeper, banker and businessman.

- A sits third to right of teacher.
- D sits second of left of G.
- G is not an immediate neighbour of teacher.
- Only one person sit between B, the shopkeeper and the teacher.
- The one who is an architect sits third to right of the shopkeeper.
- H sits between architect and engineer.
- E is not an immediate neighbour of H.
- Engineer sits third to the right of clerk.
- Only one person sits between the businessman and F.
- E is neither a businessman nor a doctor.

53. Who amongst the following is the clerk?
(a)C (b)D (c)E (d)G
54. Which of the following is true with respect to the given sitting arrangement?
(a) E is an immediate neighbour of the engineer.
(b) E is an architect.
(c) The clerk is an immediate neighbour of the banker.
(d) The teacher sits between H and the engineer.
55. What is the profession of H?
(a) Architect (b)Shopkeeper
(c)Banker (d)Teacher
56. Who sits exactly between the architect and businessman?
(a)C and H (b)Clerk
(c)Banker and shopkeeper (d)Doctor

57. Who sits immediately right of the businessman?
(a)Teacher (b)Doctor (c)Clerk (d)Banker
58. Raghav left his home for office in car. He drove 15 km straight towards North and then turned Eastwards and covered 8 km. He then turned to left and covered 1 km. He again turned left and drove for 20 km and reached office. How far and in which direction is his office from the home?
(a)20 km North –west (b)15 km North –west
(c)30 km North –west (d)25 km North
59. John is 20 years older than Steve. In 10 years, Steve's age will be half that of John. What is Steve's age now?
(a)2 (b)8 (c)10 (d)20
60. Pointing to a boy, Aruna said to Pushpa, "The mother of his father is the wife of your maternal grand father". How is Pushpa related to the boy?
(a)Sister (b)Niece
(c)Cousin sister (d)Wife
61. Which of the following pairs of numbers follow the number in the series 2, 4, 12, 24, 72, , ?
(a)144, 432 (b)288,332
(c)332, 288 (d)432, 144
62. P, Q, R, S, T and U are sitting in two rows, three in each row facing each other.
- R is second to the left of P.
 - Q and T are facing each other.
 - S and P are diagonally opposite to each other.
 - Q is not a neighbour of R.
- Which of the following are sitting in a row?
(a)P, Q, R (b)P, U, S
(c)U, T, S (d)P, T, R

Directions (Q. Nos.63 to 66) Questions are based on the following information

There are six teachers A, B, C, D, E and F in a school. Each teacher has to teach two subjects, one compulsory and the other optional.

D's optional is History, while three other have it as compulsory subject. E and F have Physics as one of their subjects. F's compulsory subject is Mathematics, which is an optional subject of both C and E.

History and English are A's subjects but in term of compulsory and optional subjects, they are reverse of D's. Chemistry is an optional subject of one of the teachers. There is only one female teacher, who has English as her compulsory subject.

63. What is C's compulsory subject?
(a)Physics (b)Chemistry
(c)English (d)History
64. Who amongst the following has Chemistry as a subject?
(a)A (b)B (c)C (d)D
65. Which of the following groups of teachers has History as the compulsory subjects?
(a) B, C and D (b) C and D
(c) A, B and C (d) A, C and D





66. Disregarding which is compulsory or optional subject who has the same two subject combination as that of F?
(a)B (b)E (c)D (d)A
67. If TRANSFER is coded as RTNAFSR, The ELEPHANT would be coded as RTNAFSRE, the
(a)LEPEHATN (b)LEPEAHTN
(c)LEPAHTN (d)LEPEAHNT
68. Which two of the following numbers comes in the next in the following sequence.
61, 57, 50, 61, 43, 36, 61,
(a)29, 61 (b)29, 20
(c) 29, 22 (d)31, 61
69. How many minimum number of colours will be required to paint all the sides of a cube without the adjacent sides having the same colours?
(a)3 (b)4 (c)5 (d)6
70. In the following sequence, which pair of numbers fill in the blanks?
1, 1, 3, 2, 8, 5, 21, 13
(a)54, 33 (b)34, 55 (c)55, 34 (d)33, 54

MATHEMATICS

71. A and B independent witness in a case. The chance that A speaks truth is x and B speaks truth is y . If A and B agree on certain statement, the probability that the statement is true, is
(a) $\frac{xy}{xy+(1-x)(1-y)}$ (b) $\frac{xy}{(1-x)(1-y)}$
(c) $\frac{(1-x)(1-y)}{xy+(1-x)(1-y)}$ (d) $\frac{x+y}{xy+(1-x)(1-y)}$
72. The harmonic mean of two numbers in 4. Their arithmetic mean A and the geometric mean G satisfy the relation $2A + G^2 = 27$, then the two numbers are
(a)4 and 2 (b)6 and 3
(c)5 and 7 (d)4 and 1
73. In an entrance test there are multiple choice questions, with four possible answers to each question of which one is correct. The probability that a student knows the answer to a question is 90%. If the student gets the correct answer to a question, then the probability that he was guessing is
(a) $\frac{37}{40}$ (b) $\frac{1}{37}$ (c) $\frac{36}{37}$ (d) $\frac{1}{9}$
74. A man is known to speak the truth 2 out of 3 times. He threw a dice cube with 1 to 6 on its faces and reports that it is 1. Then the probability that it is actually 1 is
(a) $\frac{1}{2}$ (b) $\frac{1}{7}$ (c) $\frac{2}{7}$ (d) $\frac{5}{6}$
75. Let A and B be two events such that events such that $P(\overline{A \cup B}) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P(\bar{A}) = \frac{1}{4}$, Where $\bar{A}$ stands for complement of event A. Then the event A and B are
(a) independent but not equally likely
(b) mutually exclusively and independent
(c) equally likely and mutually exclusive
(d) equally likely but not independent

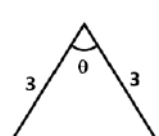
76. The mean and variance of a random variable X having binomial distribution are 4 and 2, respectively. Then $P(X = 1)$ is
(a) $\frac{1}{32}$ (b) $\frac{1}{16}$ (c) $\frac{1}{8}$ (d) $\frac{1}{4}$
77. If $\bar{x}$ is the mean of distribution of X, then usual notation $\sum_{i=1}^n f_i(x_i - \bar{x})$ is
(a) Mean deviation about mean
(b) Standard deviation
(c) 1 (d) 0
78. If E_1 and E_2 are two events associated with a random experiment such that $P(E_2) = 0.35$, $P((E_1 \text{ or } E_2)) = 0.85$ and $P((E_1 \text{ and } E_2)) = 0.15$, then $P(E_1)$ is
(a)0.25 (b)0.35 (c)0.65 (d)0.75
79. Find a matrix X such that $2A + B + X = 0$, where $A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -2 \\ 1 & 5 \end{bmatrix}$
(a) $\begin{bmatrix} 1 & 2 \\ 7 & 13 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & -2 \\ -7 & -13 \end{bmatrix}$
(c) $\begin{bmatrix} 13 & 2 \\ 7 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} -13 & -2 \\ -7 & -1 \end{bmatrix}$
80. If in a triangle ABC, the altitudes from the vertices A, B, C on opposite sides are in HP, then $\sin A, \sin B, \sin C$ are in
(a) HP
(b) Arithmetico -Geometric progression
(c) AP (d) GP
81. α, β are the roots of an equation $x^2 - 2x \cos \theta + 1 = 0$, then the equation having α^n and β^n is
(a) $x^2 - (2 \cos n \theta)x + 1 = 0$
(b) $2x^2 - (2 \cos n \theta)x - 1 = 0$
(c) $x^2 + (2 \cos n \theta)x + 1 = 0$
(d) $x^2 + (2 \cos n \theta)x - 1 = 0$
82. The equation $(x - a)^3 + (x - b)^3 + (x - c)^3 = 0$ has
(a)all three real roots
(b)one real and two imaginary roots
(c)three real roots, namely $x = a, y = b, z = c$
(d)None of the above
83. Three positive numbers whose sum is 21 are in arithmetic progression. If 2, 2, 14 are added to them respectively, then resulting numbers are in geometric progression. Then which of the following is not among the three numbers?
(a)25 (b)13 (c)1 (d)7
84. If $\sin^{-1} \frac{2a}{1+a^2} + \sin^{-1} \frac{2b}{1+b^2} = 2 \tan^{-1} n$, then
(a) $n = \frac{(a-b)}{(1+ab)}$ (b) $n = \frac{ab}{(a-b)}$
(c) $n = \frac{(a+b)}{(1-ab)}$ (d) $n = \frac{(1-ab)}{(1+ab)}$
85. The value of A that satisfies the equation $a \sin A + b \cos A - c$ is equal to
(a) $\tan^{-1} \left(\frac{a}{b} \right) \pm \cos^{-1} \left(\frac{c}{\sqrt{a^2+b^2}} \right)$
(b) $\tan^{-1} \left(\frac{c}{b} \right) \pm \sin^{-1} \left(\frac{a}{\sqrt{a^2+b^2}} \right)$
(c) $\tan^{-1} \left(\frac{a}{b} \right) \pm \sin^{-1} \left(\frac{c}{\sqrt{a^2+b^2}} \right)$
(d)None of the above



86. If $\tan x = -\frac{3}{4}$ and $\frac{3\pi}{2} < x < 2\pi$, then the value of $\sin 2x$ is
 (a) $\frac{7}{25}$ (b) $-\frac{7}{25}$
 (c) $\frac{24}{25}$ (d) $-\frac{24}{25}$
87. Find the principal value of $\cot^{-1}(-\sqrt{3})$.
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{6}$ (c) $\frac{7\pi}{6}$ (d) $\frac{5\pi}{6}$
88. If $\cos \theta = \frac{4}{5}$ and $\cos \phi = \frac{12}{13}$, with the θ and ϕ both in the fourth quadrant, the value of $\cos(\theta + \phi)$ is
 (a) $-\frac{16}{65}$ (b) $-\frac{33}{65}$ (c) $\frac{33}{65}$ (d) $\frac{16}{35}$
89. The value of $\sin 36^\circ$ is
 (a) $\frac{\sqrt{10+2\sqrt{5}}}{4}$ (b) $\frac{\sqrt{10-2\sqrt{5}}}{4}$ (c) $\frac{\sqrt{5+1}}{4}$ (d) $\frac{\sqrt{5-1}}{4}$
90. Express $(\cos 5x - \cos 7x)$
 As a product of sines or cosines or sines and cosines.
 (a) $2\cos 4x \cos x$ (b) $2\sin 4x \sin x$
 (c) $2\sin 6x \sin x$ (d) $2\cos 6x \cos x$
91. If non-zero numbers a, b, c and in A.P., then the straight line $\frac{x}{a} + \frac{y}{b} + \frac{1}{c} = 0$
 Always passes through a fixed point, then the point is
 (a) $(1, -2)$ (b) $(1, -\frac{1}{2})$ (c) $(-1, 2)$ (d) $(-1, -2)$
92. If the lines $x + (a-1)y + 1 = 0$ and $2x + a^2y - 1 = 0$ are perpendicular, then the condition satisfied by a is
 (a) $|a| = 2$ (b) $0 < a < 1$ (c) $-1 < a < 0$ (d) $a = -1$
93. If a triangle ABC, let $\angle C = \frac{\pi}{2}$.
 If r is the inradius and R is circumradius of the triangle ABC, then $2(r+R)$ equals
 (a) $a+c$ (b) $a+b+c$ (c) $a+b$ (d) $b+c$
94. If $x^2 + 3xy + 2y^2 - x - 4y - 6 = 0$ represents a pair of straight lines, their point of intersection is
 (a) $(0, 0)$ (b) $(8, 5)$ (c) $(8, -5)$ (d) $(-2, 5)$
95. The equation of the tangent line to the curve $y = 2x \sin x$ at the time $(\frac{\pi}{2}, \pi)$ is
 (a) $y = 2x + 2\pi$ (b) $y = 2x$
 (c) $y = -2x + 2\pi$ (d) $y = -2x$
96. If the graph of $y = (x-2)^2 - 3$ is shifted by 5 units up along Y-axis and 2 units to the right along the X-axis, then equation of the resultant graph is
 (a) $y = x^2 + 2$ (b) $y = (x-2)^2 + 5$
 (c) $y = (x+2)^2 + 2$ (d) $y = (x-4)^2 + 2$
97. The direction cosines of the vector $a = (-2\hat{i} + \hat{j} - 5\hat{k})$ are
 (a) $-2, 1, -5$ (b) $\frac{1}{3}, -\frac{1}{6}, -\frac{5}{6}$
 (c) $\frac{2}{\sqrt{30}}, \frac{1}{\sqrt{30}}, \frac{5}{\sqrt{30}}$ (d) $\frac{-2}{\sqrt{30}}, \frac{1}{\sqrt{30}}, \frac{-5}{\sqrt{30}}$
98. The equation of hyperbola with centre at the origin, length of the transverse axis is 6 and one focus $(0, 4)$ is
 (a) $\frac{y^2}{9} - \frac{x^2}{7} = 1$ (b) $\frac{y^2}{9} - \frac{x^2}{7} = 1$
 (c) $\frac{y^2}{7} - \frac{x^2}{9} = 1$ (d) $\frac{y^2}{7} - \frac{x^2}{9} = 1$

99. If $\vec{a}, \vec{b}, \vec{c}$ are vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$ and $|\vec{a}| = 7, |\vec{b}| = 5, |\vec{c}| = 3$, then the angle between the vectors $\vec{b}$ and $\vec{c}$ is
 (a) 60° (b) 30° (c) 45° (d) 90°
100. If $a\hat{i} + \hat{j} + k, \hat{i} + b\hat{j} + k, \hat{i} + \hat{j} + c\hat{k}$ ($a \neq b \neq c \neq 1$) are coplanar, then the value of $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is
 (a) -1 (b) $-\frac{1}{2}$ (c) $\frac{1}{2}$ (d) 1
101. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vector having magnitudes 1, 1 and 2, respectively. If $\vec{a} \times (\vec{a} \times \vec{c}) - \vec{b} = 0$, then the acute angle between $\vec{a}$ and $\vec{c}$ is
 (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{3}$ (d) None
102. Let $\vec{a}, \vec{b}, \vec{c}$ be vector such that $|\vec{a}| = 2, |\vec{b}| = 3, |\vec{c}| = 5$ and $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. The value of $\vec{a}\vec{b} + \vec{b}\vec{c} + \vec{c}\vec{a}$ is
 (a) 38 (b) -38 (c) 19 (d) -19
103. If $\vec{a} = (\hat{i} + 2\hat{j} - 3\hat{k})$ and $\vec{b} = (3\hat{i} - \hat{j} + 2\hat{k})$ then the angle between $(a+b)$ and $(a-b)$ is
 (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) $\frac{2\pi}{3}$
104. The number of elements in the power set $P(S)$ of the set $S = \{2, (1, 4)\}$ is
 (a) 2 (b) 4 (c) 8 (d) 10
105. If $(1 - x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$ then $a_0 + a_2 + a_4 + \dots + a_{2n}$ is
 (a) $\frac{3^{n+1}-1}{2}$ (b) $\frac{3^n-1}{2}$ (c) $\frac{1-3^n}{2}$ (d) $3^n + \frac{1}{2}$
106. m distinct animals of circus have to be placed in m cages, one in each cage. There are n small cages and p small animal ($n < p < m$). The large animals are so large that they do not fit in small cage. However, small animals can be put in any cage. The number of putting the animals into cage is
 (a) $\{(m-n)P_p\} \{(m-p)P_{(m-p)}\}$ (b) $(m-n)C_p$
 (c) $\{(m-n)C_p\} \{(m-p)C_{(m-p)}\}$ (d) $(m-n)P_p$
107. Let A and B two sets containing four and two elements, respectively. The number of subsets of the $A \times B$, each having at least three elements is
 (a) 270 (b) 239 (c) 219 (d) 256
108. The slope of the function

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

 (a) 1 (b) 0 (c) -1 (d) None
109. What is the largest area of an isosceles triangle with edges of length 3?

 (a) 3 (b) $\frac{3}{2}$ (c) 9 (d) $\frac{9}{2}$
110. The value of $\int_0^\pi x^3 \sin x \, dx$ is
 (a) $\pi^3 - 6\pi$ (b) $-\pi^3 - 6\pi$
 (c) $-\pi^3 + 6\pi$ (d) $\pi^3 + 6\pi$





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111. Let $f(a)$ be a polynomial of degree four, having extreme value at $x = 1$ & $x = 2$. If $\lim_{x \rightarrow 0} \left[1 + \frac{f(x)}{x^2} \right] = 3$, then $f(2)$ is
 (a) 0 (b) 4 (c) -8 (d) -4
112. The maximum value of $4 \sin^2 x + 3 \cos^2 x + \sin\left(\frac{x}{2}\right) + \cos\left(\frac{x}{2}\right)$ is
 (a) 4 (b) $3 + \sqrt{2}$ (c) 9 (d) $4 + \sqrt{2}$
113. The solution of $(e^x + 1)y \, dy = (y + 1)e^x \, dx$ is
 (a) $e^x = c(e^x + 1)(y + 1)$ (b) $e^y = e^x + y + 1$
 (c) $y = (e^x + 1)(y + 1)$ (d) None of these
114. Evaluate $\int_0^1 x(1-x)^n \, dx$
 (a) $-\frac{1}{(n+1)(n+2)}$ (b) $\frac{1}{(n+1)(n+2)}$
 (c) $(n+1)(n+2)$ (d) $(n-1)(n-2)$
115. The critical point and nature for the function $f(x, y) = x^2 - 2x + 2y^2 + 4y - 2$ is
 (a) (1, 1) maximum (b) (1, -1) maximum
 (c) (1, 1) minimum (d) (1, -1) minimum
116. If $y = \cos^2 x^2$, find $\frac{dy}{dx}$
 (a) $4x^2 \sin x^2 \cos x^2$ (b) $-4x \cos x^2 \sin x^2$
 (c) $2x \sin x^2 \cos x^2$ (d) $-2x \cos x^2 \sin x^2$
117. The derivation of $(x^3 + e^x + 3^x + \cot x)$ with respect to x is
 (a) $3x^2 + e^x + 3^x(\log 3) - \operatorname{cosec}^2 x$
 (b) $3x^2 + e^x + 3^x(\log 3) + \operatorname{cosec}^2 x$
 (c) $3x^2 + e^x + 3^x(\log 3) - \sec^2 x$
 (d) $3x^2 + e^x + 3^x(\log 3) + \sec^2 x$
118. The solution of the differential equation $\frac{dy}{dx} = e^{x+y} + x^2 e^y$ is
 (a) $e^{x-y} + \frac{x^3}{3} = C$ (b) $e^x + e^{-y} + \frac{x^3}{3} = C$
 (c) $e^x - e^{-y} = \frac{x^3}{3} + C$ (d) None of these
119. Differentiate $[-\log(\log x), x > 1]$ with respect to x .
 (a) $\frac{-1}{(x \log x)}$ (b) $\frac{1}{\log x}$ (c) $\frac{1}{x}$ (d) $x \log x$
120. Evaluate $\lim_{x \rightarrow 0} \frac{x \tan x}{(1 - \cos x)}$
 (a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) -2 (d) 2

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ANSWER KEY

1.	(b)	2.	(d)	3.	(c)	4.	(d)	5.	(c)
6.	(a)	7.	(a)	8.	(b)	9.	(d)	10.	(a)
11.	(c)	12.	(a)	13.	(a)	14.	(a)	15.	(c)
16.	(b)	17.	(c)	18.	(b)	19.	(c)	20.	(d)
21.	(c)	22.	(a)	23.	(d)	24.	(d)	25.	(c)
26.	(a)	27.	(c)	28.	(b)	29.	(d)	30.	(d)
31.	(d)	32.	(b)	33.	(c)	34.	(d)	35.	(b)
36.	(b)	37.	(a)	38.	(b)	39.	(d)	40.	(b)
41.	(a)	42.	(c)	43.	(d)	44.	(b)	45.	(b)
46.	(d)	47.	(a)	48.	(c)	49.	(a)	50.	(b)
51.	(b)	52.	(d)	53.	(d)	54.	(c)	55.	(d)
56.	(d)	57.	(c)	58.	(a)	59.	(c)	60.	(c)
61.	(a)	62.	(d)	63.	(d)	64.	(b)	65.	(c)
66.	(b)	67.	(b)	68.	(c)	69.	(a)	70.	(c)
71.	(a)	72.	(b)	73.	(b)	74.	(c)	75.	(a)
76.	(a)	77.	(d)	78.	(c)	79.	(b)	80.	(c)
81.	(a)	82.	(b)	83.	(b)	84.	(c)	85.	(d)
86.	(d)	87.	(d)	88.	(c)	89.	(b)	90.	(c)
91.	(a)	92.	(d)	93.	(c)	94.	(c)	95.	(b)
96.	(d)	97.	(d)	98.	(b)	99.	(a)	100.	(d)
101.	(b)	102.	(d)	103.	(c)	104.	(b)	105.	(a)
106.	(a)	107.	(c)	108.	(b)	109.	(d)	110.	(a)
111.	(a)	112.	(d)	113.	(d)	114.	(a)	115.	(d)
116.	(b)	117.	(a)	118.	(b)	119.	(a)	120.	(d)



SOLUTION

1.	(b)	2.	(d)	3.	(c)	4.	(d)	5.	(c)
6.	(a)	7.	(a)	8.	(b)	9.	(d)	10.	(a)
11.	(c)	12.	(a)	13.	(a)	14.	(a)	15.	(c)
16.	(b)	17.	(c)	18.	(b)	19.	(c)	20.	(d)

COMPUTER

21. (c) $A, A' = 0$

So, option (a) is wrong

$A + AB$

$A(1+B) = A$

So, option (b) is wrong

$A + A'B$

$(A+A')(A+B)$

$1(A+B) = A+B$

So, option (c) is right.

$A(A+B)$

$A.A+B = A+B$

So, option (d) is wrong.

22. (a)

The binary number + 1001.11 is represented with an 8-bit fraction and 6-bit exponent as follows

Fraction Exponent

01001110 000100

The fraction has a 0 in the left most position to denote positive.

The binary point of the fraction follows the sign bit but is not shown in the register. The exponent has the equivalent binary number +4. The floating point number is equivalent to

$m \times 2^e = +(.1001110)_2 \times 2^{+4}$

23. (d)

$AB + A'C + BC = AB + A'C + BC(A+A')$

[by Complementary law]

$= AB + A'C + ABC + A'BC$

$= AB + ABC + A'C + A'BC$

[By Associative law]

$= AB(1+C) + A'C(1+B)$ [by Absorption law]

$= AB + A'C$

24. (*) Hit Ratio = 15%

Cache memory time = 10 microseconds

Memory access time = 10 milliseconds

Effective Memory Access Time (EAT) = Hit Ratio

(Cache memory time + Memory access time) + (sign hit)*

(Cache memory time + 2* Memory access time)

$EAT = 0.15(0.01) + 10 + (1-0.15)(0.01 + 2 \times 10)$

$= (0.15)(10.01) + (0.85)(20.01)$

$= 1.5015 + 17.0085 = 18.51 \text{ milliseconds}$

None option is correct.

25. (c)

$(147)_{10}$ convert into $(?)_2$

2	147	
2	73	1
2	36	1
2	18	0
2	9	0
2	4	1
2	2	0
	1	0

$(147)_{10} = (10010011)_2$

First bit (the left most) indicates the sign

1 = negative, 0 = Positive

Positive binary computer representation on 12 bit if needed, add extra 0, in front of the base 2 member.

So, $(147)_{10} = 000010010011$

For 1's complement, replace all the bits on 0 with 1s and all the bits set on 1 with 0s.

1's complement of

$(-147) = 111101101100$

For 2's complement, add 1 to the number of 1's complement

2's complement of (-147)

111101101100

+ 1

111101101101

26. (a) RAM.

27. (c)

Changing (123) base 5 into base 10.

$= 1 \times 5^2 + 2 \times 5^1 + 3 \times 5^0$

$= 1 \times 25 + 2 \times 5 + 3 \times 1^0$

$= 25 + 10 + 3 = 38$ (i)

Changing x_8 base in decimal

$= x \cdot y^1 + 8 \cdot y^0$

$= xy + 8$ (ii)

From Eqs. (i) and (ii), we get

$xy + 8 = 38$

$\Rightarrow xy = 30$

So, possible combinations = (1, 30), (2, 15), (3, 10)

But we have 8 present in x_8 , so base $y > 8$

Therefore, total solution = 3

28. (b)

The range of numbers that can be represented by n-bits in 2's complement form is $(-2)^{n-1}$ to $(2^{n-1}) - 1$.

Hence, the smallest number is $(-2)^{n-1}$

$= (-2)^{8-1} = (-2)^7 = -128$

29. (d)

A functionally complete set of Boolean function consists of a set of Boolean functions from which you can construct all Boolean function. {AND, NOT}, {OR, NOT}, {NAND}, (NOR) are four functionally complete sets.

30. (d)

For n-Boolean variables, there are 2^n possible Boolean inputs. Each input can generate either True or False as the output. Hence, total number of different Boolean function = 2^{2^n} .

REASONING

31. (d) If all the given statement are false, then the information will be as follows

(i) $S > 40$ years

(ii) S is 1,2,3,4 or 5 years older than M and $M \neq 38$ years

(iii) $M \leq 39$ years

According to the information and from options, Ms age can be of 36 or 37 or 39 years.

And S's age can be of 40 or 41 or 42 or 43 or 44 years.

Hence, exact age of M and S cannot be determined.



32. (b) Let largest number is P
and any two chosen numbers x and y .
According to the question $P + x - y$ is divisible by 10.

So,
 P is largest then
 $P + x - y \geq 10$
 $x - y \leq P$

These inequalities can be written in equation form and

$$P + x - y = 10$$

$$x - y = P$$

On subtracting eq. (iv) by (iii), we get

$$P + x - y - x + y = 10 - P$$

$$\Rightarrow P = 10 - P$$

$$\Rightarrow 2P = 10$$

$$P = \frac{10}{2} = 5$$

33. (c) According to the given information,

$$A < B, C < D, B < C \text{ and } A < E$$

$$A < B < C < D \text{ and } A < E$$

$\therefore$ E's position can't be determined.

Hence, most intelligent child is either D or E.

34. (d) There are three number of ways by which we can form the committee.

(i) 3 Males and 2 Females

(ii) 4 Males and 1 Female

(iii) 5 Males + 0 Female

Hence, total number of ways

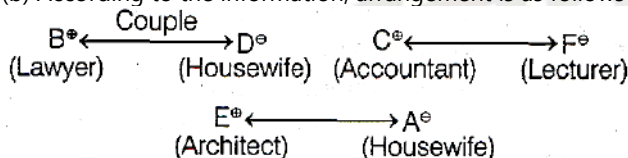
$$= {}^7C_3 {}^6C_2 + {}^7C_4 {}^6C_1 + {}^7C_5 {}^6C_0$$

$$= \frac{7 \times 6 \times 5}{3 \times 2 \times 1} \times \frac{6 \times 5}{2 \times 1} + \frac{7 \times 6 \times 5}{3 \times 2 \times 1} \times \frac{6}{1} + \frac{7 \times 6}{2 \times 1} \times 1$$

$$= 525 + 210 + 21 = 756$$

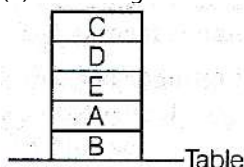
Hence, this can be done by 756 ways.

35. (b) According to the information, arrangement is as follows



E is an architect.

36. (b) According to the information, the arrangement is as follows



Hence, book B touches the surface of the table.

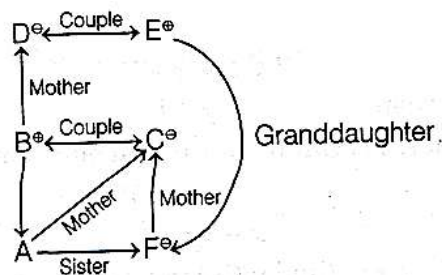
37. (a) Decimal 99 is represented in different number system in the series.

99	90	83	78	71	69
Base = 10	Base = 11	Base = 12	Base = 13	Base = 14	Base = 15
$= 9 \times 10 + 9$	$= 9 \times 11 + 0$	$= 8 \times 12 + 3$	$= 7 \times 13 + 8$	$= 7 \times 14 + 1$	$= 6 \times 15 + 9$
$= (99)_{10}$	$= (90)_{11}$	$= (83)_{12}$	$= (78)_{13}$	$= (71)_{14}$	$= (69)_{15}$

Hence, (71, 69) is the correct answer.

Sol. (Q. Nos. 38-40)

According to the information, arrangement is as follows



38. (b) C is the mother of A.

39. (d) As gender of A is not known, therefore none of the option is correct.

40. (b) DE and BC are the couples in the family.

41. (a) The pattern of the series is as follows

336	210	120	60	24	6
$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
$= 7^3 - 7$	$= 6^3 - 6$	$= 5^3 - 5$	$= 4^3 - 4$	$= 3^3 - 3$	$= 2^3 - 2$

42. (c) Three day before Friday is Tuesday.

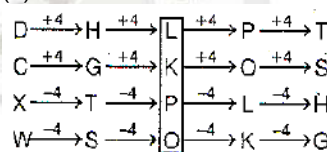
$\therefore$ The day after the day after tomorrow = Tuesday

$\therefore$ The day after the day tomorrow = Monday

$\therefore$ The day on tomorrow = Sunday

$\therefore$ Today = Saturday

43. (d)



So the missing term = LKPO

44. (b) According to the information, arrangement is as follow

Profession	Language	Religion
Lawyer	Bengali	Christian/Muslim
Lawyer	Hindi	Muslim
Doctor	Bengali	Muslim/Christian
Doctor	Hindi	Christian

$\therefore$ Lawyer is either Christian or Muslim, he will definitely speaks Bengali.

Hence, option (b) is the correct answer.

45. (b) From all the conditions given in the questions, Option (a) is not possible as Harish, Javed and Laxman cannot ride together given in condition (iii).

Option (c) is not possible as in condition (v), in Kumar or Laxman one person is compulsory. No one is given in option (c).

Option (d) is not possible as Kumar and Laxman cannot ride together.

Hence, option (b) is correct which fulfills all the conditions.

46. (d) If Javed rides then either Kumar or Mohan will definitely ride by condition (iv).

But, Javed and Mohan is already given. Thus, Kumar will not ride.

But in condition (v), one person from Kumar or Laxman will definitely ride. Hence, Laxman must ride.



47. (a) If Feroz and Harish both ride, then Gautam must ride. By condition (ii), Javed will not ride.
By condition (vi), either Kumar or Mohan will ride and by condition (v), either Kumar or Laxman is compulsory.
If Mohan rides, then Laxman will ride.
And, if Kumar rides then Laxman cannot ride.
Hence, Feroz, Harish, Gautam, Mohan and Laxman can ride.

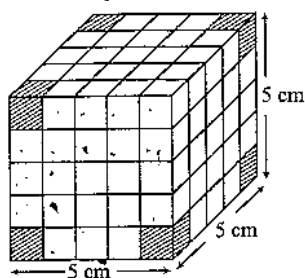
48. (c)

	A	E	F	G	H	I	B
O		J	K	L	M	N	P
Q		R	S	T	U	V	W
X		Y	Z	A'	B'	C'	D'
	D	E'	F'	G'	H'	I'	C

Total number of squares

= AEJO, EFJK, FGLK, GHML, HINM, IBPN, JRQO, JKSR, KLTS, LMUT, MNVU, NPWV, QRYX, RSZY, STAZ, TUB' A', UVCB', VWD'C', DEYX, YZFE', ZA'GF', AB'H'G', B'C'H', CD'CI', AFSQ, EGTR, FHUS, GIVT, HBWU, OKZX, JLAY, KMBZ, LNC'A', MPDB', QSFD, RTGE', SUHF', TVI'G', UWCH', AGAX, EHBY, FICZ, GBD'A', OLGD, JMHE', KNIF', LPCG', AHHD, EII', FBCF'.

49. (a) Four smaller cubes at the corners of the larger cube are visible only on three sides when cube placed on a table.



50. (b) Let number of breads available be 'x'.

According to question,

Number of breads robbed by 1st thief,

$$= \frac{x}{2} + \frac{1}{2} = \frac{x+1}{2}$$

Now, number of breads left after 1st robbery,

$$= x - \frac{x+1}{2} = \frac{x-1}{2}$$

Again, number of breads robbed by 2nd thief,

$$= \frac{1}{2} \left(\frac{x-1}{2} \right) + \frac{1}{2}$$

$$= \frac{x-1}{4} + \frac{1}{2} = \frac{x+1}{4}$$

So, number of bread left after 2nd robbery,

$$= \left(\frac{x-1}{2} \right) - \left(\frac{x+1}{4} \right) = \frac{2x-2-x-1}{4} = \frac{x-3}{4}$$

Now, number of breads robbed by 3rd thief,

$$= \frac{1}{2} \left(\frac{x-3}{4} \right) + \frac{1}{2} = \frac{x-3}{8} + \frac{1}{2} = \frac{x+1}{8}$$

Hence, number of breads left after 3rd robbery,

$$= \left(\frac{x-3}{4} \right) - \left(\frac{x+1}{8} \right) = \frac{x-7}{8}$$

Now,

$$\frac{x-7}{8} = 3$$

$$x = 31$$

$$\Rightarrow x = 31$$

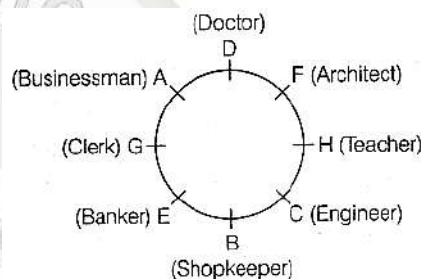
Therefore, total number of breads present initially is 31.

51. (b) Caterpillar crawls upward $(5 - 4)$, 1 inch in a day.
Upto 70th day, caterpillar crawls upward 70 inches.
On 71st day, caterpillar crawls upward 75 cm at the end of the day.

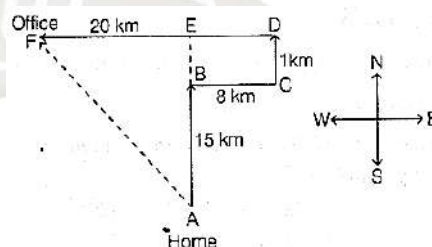
(d)	Investment	Amount he get
In	1990 →	₹5000
In	1995 →	₹5000 + ₹10000
In	2000 →	₹5000 + ₹30000
In	2005 →	₹5000 + ₹70000
In	2010 →	0 + ₹150000

He received ₹ 1,50,000 in 2010.

Sol. (Q. Nos. 53-57) According to the arrangement is as follows



53. (d) G is the clerk.
54. (c) The clerk is an immediate neighbour of the banker, is true.
55. (d) The profession of H is teacher.
56. (d) Doctor sits between the architect and businessman.
57. (c) Clerk sits immediately right of the businessman.
58. (a)



$$\begin{aligned} AF &= \sqrt{(FE)^2 + (AE)^2} \\ \Rightarrow AF &= \sqrt{(20-8)^2 + (15+1)^2} \\ &= \sqrt{144 + 256} \\ &= \sqrt{400} \\ &= 20 \text{ km} \end{aligned}$$

Raghav's office is 20 km away from his home and in North-West direction.

59. (c) Let the age of Steve be 'S' and John be 'J'.

According to the question,

$$\text{And, } J = S + 20$$

$$\Rightarrow S + 10 = \frac{J+10}{2}$$

$$\Rightarrow 2S + 20 = J + 10$$

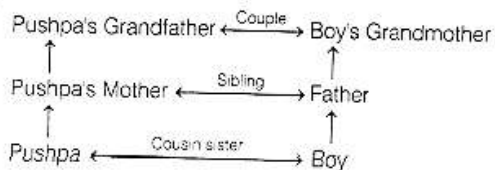
$$\Rightarrow 2S + 10 = S + 20$$

$$S = 10$$

Present age of Steve is 10 years.

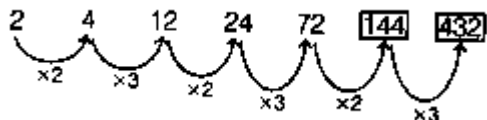


60. (c)

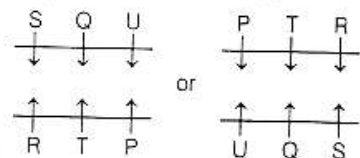


Hence, Pushpa is the cousin sister of boy in the photograph.

61. (a) The pattern of the series is as follows



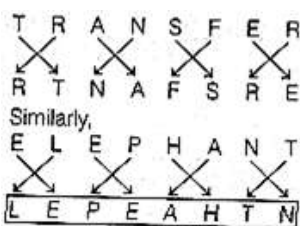
62. (a) The arrangement is as follows



Sol. (Q. Nos. 63-66) According to the information, arrangement is as follows :

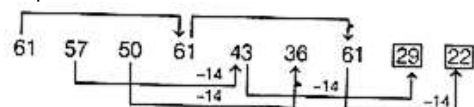
	Compulsory Subject	Optional Subject
A	History	English
B	History	Chemistry
C	History	Mathematics
D	English	History
E	Physics	Mathematics
F	Mathematics	Physics

63. (d) History is C's compulsory subject.
 64. (b) B has Chemistry as a subject.
 65. (c) Group 'A, B and C' has history as the compulsory subject.
 66. (b) E has the same two subject combination as that of F.
 67. (b) As,

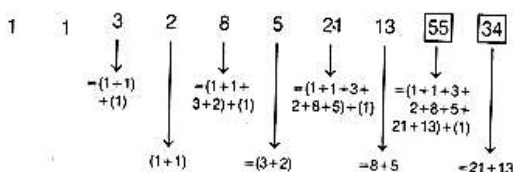


68. (c)

The pattern of the series is as follows

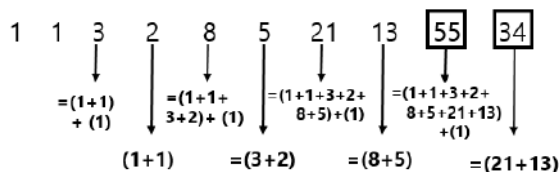


69. (a) Three different colours are required to point all the sides of a cube so that no two adjacent sides having the same colours. Opposite faces have same colour.
 70. The pattern of the series is as follows



MATHS

71.



72. (b) Let the two number be a and b.

According to the question, we have

$$\frac{2ab}{a+b} = 4$$

$$\text{and } 2A + G^2 = 27$$

$$\left[\because A = \frac{a+b}{2}, G = \right]$$

$$\text{so } ab = 2(a+b) \quad \dots(i)$$

$$\text{and } a + b + ab = 27 \quad \dots(ii)$$

$$\Rightarrow a + b + 2(a+b) = 27$$

$$\Rightarrow 3(a+b) = 27$$

$$\Rightarrow a+b = 9 \quad \dots(iii)$$

$\Rightarrow$ From Eqs. (i) and (iii), we have

$$\Rightarrow ab = 18 \quad \dots(iv)$$

$\Rightarrow$ Putting, $b = \frac{18}{a}$ from Eqs. (iv) in Eqs. (iii), we get

$$\Rightarrow a + \frac{18}{a} = 9$$

$$\Rightarrow a^2 - 9a + 18 = 0$$

$$\Rightarrow (a-3)(a-6) = 0$$

$$\Rightarrow a = 3, 6$$

$\therefore$ Required numbers are 3 and 6.

73. (b) Let E_1 be the event that student knows the answer and E_2 be the event that student guesses the answer. A be the event that he answered correctly.

$$P(E_1) = 0.90, P(E_2) = 0.10$$

$$P\left(\frac{A}{E_1}\right) = 1, P\left(\frac{A}{E_2}\right) = 0.25$$

Required probability,

$$P\left(\frac{E_2}{A}\right) = \frac{P(E_2)P\left(\frac{A}{E_2}\right)}{P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right)}$$

$$= \frac{0.10 \times 0.25}{0.90 \times 1 + 0.10 \times 0.25}$$

$$= \frac{25 \times 10}{90 \times 100 + 10 \times 25} = \frac{25}{900 + 25} = \frac{1}{36 + 1} = \frac{1}{37}$$

74. (c) Let E_1 and E_2 be the events that man speaks truth and lie, respectively. Also, let A be the event that 1 appear on dice.

Now, according to the questions, we know that

$$P(E_1) = \frac{2}{3}, P(E_2) = \frac{1}{3}$$

$$P\left(\frac{A}{E_1}\right) = \frac{1}{6}, P\left(\frac{A}{E_2}\right) = \frac{5}{6}$$

$\therefore$ Required probability,

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1)P\left(\frac{A}{E_1}\right)}{P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right)}$$

$$= \frac{\frac{2}{3} \times \frac{1}{6}}{\frac{2}{3} \times \frac{1}{6} + \frac{1}{3} \times \frac{5}{6}} = \frac{2}{2+5} = \frac{2}{7}$$

75. We have,

$$P(\overline{A \cup B}) = \frac{1}{6}$$

$$P(A \cup B) = \frac{1}{4}$$

$$P(\overline{A}) = \frac{1}{4}$$



76. (a) According to the questions, we have

$$\text{Mean} = np = 4 \text{ and variance} = npq = 2$$

$$\text{So } 4q = 2$$

$$\Rightarrow np = 4$$

$$\text{Now, } p = 1 - q = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\text{And } np = 4$$

$$\Rightarrow n \times \frac{1}{2} = 4$$

$$\Rightarrow n = 8$$

$$\text{So } n = 8, p = \frac{1}{2}, q = \frac{1}{2}$$

Now by binomial distribution we know that

$$P(X = r) = {}^nC_r p^r q^{n-r}$$

$$P(X=1) = {}^8C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{8-1} = 8 \times \left(\frac{1}{2}\right)^8 = \frac{1}{32}$$

77. (d) By property of arithmetic mean, the algebraic sum of all the deviation taken about the arithmetic mean is always zero.

$$\text{So } \sum_{i=1}^n (x_i - \bar{x}) = 0$$

78. (c) We have

$$P(E_2) = 0.35$$

$$P(E_1 \text{ or } E_2) = 0.85$$

$$P(E_1 \text{ and } E_2) = 0.15$$

We know that,

$$P(E_1 \text{ or } E_2) = P(E_1) + P(E_2) - P(E_1 \text{ and } E_2)$$

$$\Rightarrow 0.85 = P(E_1) + 0.35 - 0.15$$

$$\Rightarrow P(E_1) = 0.85 - 0.35 + 0.15 = 0.65$$

79. (b) we have,

$$A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 3 & -2 \\ 1 & 5 \end{bmatrix}$$

Now,

$$2A + B + X = 0$$

$$\Rightarrow X = -(2A+B)$$

$$= -\left(2 \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 3 & -2 \\ 1 & 5 \end{bmatrix}\right)$$

$$= -\left(2 \begin{bmatrix} -2 & 4 \\ 6 & 8 \end{bmatrix} + \begin{bmatrix} 3 & -2 \\ 1 & 5 \end{bmatrix}\right)$$

$$= -\begin{bmatrix} 1 & 2 \\ 7 & 13 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ -7 & -13 \end{bmatrix}$$

80. (c) Let, the altitude from A, B, C are h_1, h_2, h_3

Respectively,

We know that,

$$\text{Area of triangle, } \Delta = \frac{1}{2}ah_1 = \frac{1}{2}bh_2 = \frac{1}{2}ch_3$$

$$\therefore a = \frac{2\Delta}{h_1}, b = \frac{2\Delta}{h_2}, c = \frac{2\Delta}{h_3}$$

Since, h_1, h_2, h_3 are in HP.

$\Rightarrow a, b, c$ are in AP.

$\Rightarrow k \sin A, k \sin B, k \sin C$ are in AP

$\Rightarrow \sin A, \sin B, \sin C$ are in AP.

81. (a) We have,

$$X^2 - 2x \cos \theta + 1 = 0$$

$$\Rightarrow \frac{2 \cos \theta \pm \sqrt{4 \cos^2 \theta - 4}}{2} = \frac{2 \cos \theta \pm 2 \sin \theta}{2}$$

$$= \cos \theta \pm i \sin \theta$$

$$\text{so } \alpha = \cos \theta + i \sin \theta$$

$$\text{and } \beta = \cos \theta - i \sin \theta$$

$$\text{so } \alpha^n = (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

$$\text{and } \beta^n = (\cos \theta - i \sin \theta)^n = \cos n\theta - i \sin n\theta$$

[using De-Moivre's Theorem]

$$\text{Now, } \alpha^n + \beta^n = 2 \cos n\theta \text{ and } \alpha^n \cdot \beta^n = 1$$

So Equation having α^n and β^n as roots is

$$x^2 - (\alpha^n + \beta^n)x + \alpha^n \beta^n = 0$$

$$\Rightarrow x^2 - 2 \cos n\theta x + 1 = 0$$

82. (b) Let

$$f(x) = (x-a)^3 + (x-b)^3 + (x-c)^3$$

$$f'(x) = 3(x-a)^2 + 3(x-b)^2 + 3(x-c)^2$$

$$= 3[(x-a)^2 + (x-b)^2 + (x-c)^2]$$

$$\text{So } f'(x) > 0 \forall x \in \mathbb{R}$$

$$\text{So } f'(x) = 0 \text{ has no real roots.}$$

Hence, $f(x) = 0$ has two imaginary and one real root.

83. (b) Let the three numbers in AP are $a-d, a, a+d$.

It is given that

$$a-d + a + a+d = 21$$

$$\Rightarrow 3a = 21$$

$$\Rightarrow a = 7$$

So numbers are $7-d, 7, 7+d$

Again, it is given that

$$7-d + 2, 7+2, 7+d + 14 \text{ are in G.P.}$$

$$\Rightarrow 9-d, 9, 21+d \text{ are in G.P.}$$

$$\Rightarrow 9^2 = (9-d)(21+d)$$

$$\Rightarrow 81 = 189 + 9d - 21d - d^2$$

$$\Rightarrow d^2 + 12d - 108 = 0$$

$$\Rightarrow (d-6)(d+18) = 0$$

$$\Rightarrow d = 6, -18$$

Putting $d = 6$ in $7-d, 7, 7+d$, we get 1, 7, 13

So 25 is not among the three numbers.

84. (c) We have,

$$\sin^{-1} \frac{2a}{1+a^2} + \sin^{-1} \frac{2b}{1+b^2} = 2 \tan^{-1} n$$

$$= 2 \tan^{-1} a + 2 \tan^{-1} b = 2 \tan^{-1} n$$

$$[\because 2 \tan^{-1} x = \sin^{-1} \frac{2x}{1+x^2}]$$

$$= \tan^{-1} a + \tan^{-1} b = \tan^{-1} n$$

$$= \tan^{-1} \left(\frac{a+b}{1-ab} \right) = \tan^{-1} n$$

$$= \frac{a+b}{1-ab} = n$$

85. (d) Given,

$$a \sin A + b \cos A - c = 0 \dots\dots(i)$$

$$\text{Let } a = r \cos \alpha$$

$$b = r \sin \alpha$$

$$= r = \sqrt{a^2 + b^2}$$

$$\tan \alpha = \frac{b}{a}$$

$$= \alpha = \tan^{-1} \left(\frac{b}{a} \right)$$

Substituting these values in Eq. (i) we get

$$r \cos \alpha \sin A + r \sin \alpha \cos A = c$$

$$r \sin(A + \alpha) = c$$

$$\sin(A + \alpha) = \frac{c}{\sqrt{a^2 + b^2}}$$

$$A + \alpha = \sin^{-1} \left(\frac{c}{\sqrt{a^2 + b^2}} \right)$$

$$A = \sin^{-1} \left(\frac{c}{\sqrt{a^2 + b^2}} \right) - \tan^{-1} \left(\frac{b}{a} \right)$$

86. (d) Given that,

$$\tan x = -\frac{3}{4} \text{ and } \frac{3\pi}{2} < x < 2\pi$$

Since, x lies in the fourth quadrant

$$\text{Then } \sin x = -\frac{3}{5} \text{ and } \cos x = \frac{4}{5}$$

As we know that,

$$\sin 2x = 2 \sin x \cos x$$



$$= -x \left(-\frac{3}{5}\right) \times \frac{4}{5} = -\frac{24}{25}$$

87. (d) Since, $\cot^{-1}(-x) = \pi - \cot^{-1}(x)$

$$\therefore \cot^{-1}(-\sqrt{3}) = \pi - \cot^{-1}(\sqrt{3})$$

$$= \pi - \cot^{-1}\left(\cot\frac{\pi}{6}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

88. (c) Given, $\cos \theta = \frac{4}{5}$, $\cos \phi = \frac{12}{13}$

It is given that θ and ϕ both lies in fourth quadrant.

$$\therefore \sin \theta = -\frac{3}{5} \text{ and } \sin \phi = -\frac{5}{13}$$

$$\text{Now, } \cos(\theta + \phi) = \cos \theta \cos \phi - \sin \theta \sin \phi$$

$$= \frac{4}{5} \times \frac{12}{13} - \left(-\frac{3}{5}\right) \left(-\frac{5}{13}\right) = \frac{48}{65} - \frac{15}{65} = \frac{33}{65}$$

89. (b) Since, $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$

As we know that,

$$\cos 36^\circ = 1 - 2 \sin^2 18^\circ$$

$$= 1 - 2 \left(\frac{\sqrt{5}-1}{4}\right)^2 = 1 - 2 \left(\frac{5+1-2\sqrt{5}}{16}\right)$$

$$= 1 - \left(\frac{6-2\sqrt{5}}{8}\right) = \frac{8-6+2\sqrt{5}}{16}$$

$$= \frac{\sqrt{5}+1}{4}$$

$$\sin^2 36^\circ = 1 - \cos^2 36^\circ$$

$$1 - \left(\frac{5+1+2\sqrt{5}}{16}\right)$$

$$= \frac{16-6-2\sqrt{5}}{16} = \frac{10-2\sqrt{5}}{16}$$

$$\sin 36^\circ = \sqrt{\frac{10-2\sqrt{5}}{16}} = \frac{\sqrt{10-2\sqrt{5}}}{4}$$

- 90.(c) Given that

$$\cos 5x - \cos 7x$$

We know that,

$$\cos C - \cos D = 2 \sin\left(\frac{C+D}{2}\right) \sin\left(\frac{D-C}{2}\right)$$

$$\cos 5x - \cos 7x = 2 \sin\left(\frac{5x+7x}{2}\right) \sin\left(\frac{7x-5x}{2}\right)$$

$$= 2 \sin\left(\frac{12x}{2}\right) \sin\left(\frac{2x}{2}\right)$$

$$= 2 \sin 6x \sin x$$

91. (*) Given, numbers a, b, c are in AP

$$\therefore 2b = a + c \quad \dots(i)$$

Now, the given straight line,

$$\frac{x}{a} + \frac{y}{b} + \frac{1}{c} = 0$$

$$= \frac{x}{a} + \frac{y}{b} = -\frac{1}{c}$$

$$= \frac{c}{\left(-\frac{a}{c}\right)} + \frac{y}{\left(-\frac{b}{c}\right)} = 1 \quad \dots(ii)$$

By using condition (i), we cannot determine that line (ii) passes through from which point. So we can say that given condition for question is wrong for concluding a definite result.

92. (d) Given equations of line are

$$x + (a-1)y + 1 = 0 \quad \dots(i)$$

$$\text{and } 2x + a^2y - 1 = 0 \quad \dots(ii)$$

since, line (i) and line (ii) are perpendicular

$$\therefore \text{slope of line 1} \times$$

$$\text{slope of line 2} = -1$$

$$\therefore \text{slope of line } x + (a-1)y + 1 = 0$$

$$m_1 = \frac{-1}{a-1}$$

$$\text{slope of line } 2x + a^2y - 1 = 0$$

$$m_2 = \frac{-2}{a^2}$$

$$\therefore m_1 m_2 = -1$$

$$\frac{1}{(a-1)} \times \frac{2}{a^2} = -1$$

$$= 2 = -a^2(a-1)$$

$$= 2 = -a^3 + a^2$$

$$= a^3 + a^2 - 2 = 0 \quad \dots(iii)$$

On solving Equation (iii) we get

$$a = -1$$

93. (c) Given,

$$\angle C = \frac{\pi}{2}$$

$$\therefore R = \frac{c}{2 \sin C} = \frac{c}{2 \sin 90^\circ} = \frac{c}{2}$$

$$\text{And } r = (s-c) \tan \frac{C}{2} = (s-c) \tan 45^\circ$$

$$r = (s-c)$$

$$\therefore 2(r+R) = 2\left(\frac{c}{2} + s - c\right)$$

$$= 2s - c$$

$$= a + b + c - c$$

$$= a + b$$

94. (c) As we know that the pair of straight line

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ intersect each other at}$$

$$\left(\frac{hf-bg}{ab-h^2}, \frac{hg-af}{ab-h^2}\right)$$

Therefore, points of intersection of pair of straight line

$$x^2 + 3xy + 2y^2 - x - 4y - 6 = 0 \text{ is}$$

$$\left\{\frac{\frac{3}{2} \times (-2) - (2) \left(-\frac{1}{2}\right)}{(1 \times 2) - \left(\frac{3}{2}\right)^2}, \frac{\frac{3}{2} \times \left(-\frac{1}{2}\right) - 1 \times (-2)}{1 \times 2 - \left(\frac{3}{2}\right)^2}\right\}$$

$$= \frac{-3+1}{2-\frac{9}{4}}, \frac{-\frac{3}{4}+2}{2-\frac{9}{4}}$$

$$= \left(-\frac{2}{\frac{1}{4}}, \frac{\frac{5}{4}}{\frac{1}{4}}\right) = (8, -5)$$

95. (b) Given equation of curve is

$$y = 2x \sin x$$

$$\text{slope at point } \left(\frac{\pi}{2}, \pi\right)$$

$$\left(\frac{dy}{dx}\right)_{\left(\frac{\pi}{2}, \pi\right)} = 2 \sin x + 2x \cos x$$

$$= 2 \sin \frac{\pi}{2} + 2 \times 0$$

$$= 2 + 0 = 2$$

$\therefore$ Equation of tangent at point $\left(\frac{\pi}{2}, \pi\right)$ is

$$y - \pi = \left(\frac{dy}{dx}\right)_{\left(\frac{\pi}{2}, \pi\right)} \left(x - \frac{\pi}{2}\right)$$

$$y - \pi = 2x - \pi$$

$$y = 2x$$

96. (d) We have,

$$y = (x-2)^2 - 3 \quad \dots(i)$$

Since, y is shifted up along Y-axis by 5 units and x is shifted along right by 2 units.

$\therefore$ Put $y = y - 5$ and $x = x - 2$ in Eq. (i), we get

$$y - 5 = (x-2-2)^2 - 3$$

$$= y - 5 = (x-4)^2 - 3$$

$$= y = (x-4)^2 + 2$$

97. (d) We have,

$$\vec{a} = -2\hat{i} + \hat{j} - 5\hat{k}$$

$$\therefore |\vec{a}| = \sqrt{(-2)^2 + (1)^2 + (-5)^2}$$

$$= \sqrt{4 + 1 + 25} = \sqrt{30}$$

$$\therefore \text{Unit vector, } \hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

$$= \frac{-2\hat{i} + \hat{j} - 5\hat{k}}{\sqrt{30}} = \frac{-2}{\sqrt{30}} \hat{i} + \frac{1}{\sqrt{30}} \hat{j} - \frac{5}{\sqrt{30}} \hat{k}$$

$$\text{DC's of } \vec{a} = \frac{-2}{\sqrt{30}}, \frac{1}{\sqrt{30}}, \frac{-5}{\sqrt{30}}$$

98. (b) We have,

Focus = (0, 4) and length of transverse axis = 6

$$\therefore be = -4 \text{ and } 2b = 6$$



$$= be = 4 \text{ and } b = 3$$

$$= e = 4/3 \text{ and } b = 3$$

$$\text{Also } e^2 = 1 + \frac{a^2}{b^2}$$

$$= \frac{16}{9} = 1 + \frac{a^2}{9}$$

$$= a^2 = 7$$

$\therefore$ Equation of hyperbola is

$$= \frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$$

$$= \frac{x^2}{7} - \frac{y^2}{9} = -1$$

$$= \frac{y^2}{9} - \frac{x^2}{7} = 1$$

99. (a) We have,

$$|\vec{a}| = 7, |\vec{b}| = 5, |\vec{c}| = 3$$

$$\text{Now, } \vec{a} + \vec{b} + \vec{c} = 0$$

$$= \vec{b} + \vec{c} = -\vec{a}$$

$$= |\vec{b} + \vec{c}| = |\vec{a}|$$

$$= |\vec{b} + \vec{c}|^2 = |\vec{a}|^2$$

$$= (\vec{b} + \vec{c}) \cdot (\vec{b} + \vec{c}) = |\vec{a}|^2$$

$$= |\vec{b}|^2 + |\vec{c}|^2 + 2\vec{b} \cdot \vec{c} = |\vec{a}|^2$$

$$25 + 9 + 2|\vec{b}||\vec{c}|\cos\theta = 49$$

Where θ is angle between $\vec{b}$ and $\vec{c}$

$$= 2 \times 5 \times 3 \cos\theta = 15$$

$$\cos\theta = \frac{1}{2}$$

$$\theta = 60^\circ$$

100. (d) If $a\hat{i} + \hat{j} + \hat{k}, \hat{i} + b\hat{j} + \hat{k}, \hat{i} + \hat{j} + c\hat{k}$ ($a \neq b \neq c \neq 1$)

Coplanar, then

$$\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$$

$$\begin{vmatrix} a & 1-a & 0 \\ 1 & b-1 & 1-b \\ 1 & 0 & c-1 \end{vmatrix} = 0 \left(\begin{matrix} C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 - C_1 \end{matrix} \right)$$

$$= (1-a)(1-b)(1-c) \begin{vmatrix} \frac{a}{1-a} & 1 & 0 \\ \frac{a}{1-b} & -1 & 1 \\ \frac{1}{1-c} & 0 & -1 \end{vmatrix} = 0$$

$$= \frac{a}{1-a}(1-0) - \frac{1}{1-b}(-1-0) + \frac{1}{1-c}(1-1)$$

$$= \frac{a}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$$

$$= \frac{a+1-a}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$$

$$\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$$

101. (b) Given, $|a| = |b| = 1, |c| = 2$

$$a \times (a \times c) - b = 0$$

$$(a \cdot c)a - (a \cdot a)c = b$$

$$(a \cdot c)a - |a|^2 c = b$$

$$|(a \cdot c)a - c|^2 = |b|^2$$

$$\Rightarrow |a \cdot c|^2 |a|^2 + |c|^2 - 2(a \cdot c)(a \cdot c) = 1$$

$$\Rightarrow |a \cdot c|^2 + 4 - 2|a \cdot c|^2 = 1 \Rightarrow 4 - |a \cdot c|^2 = 1$$

$$\Rightarrow 4 - |a|^2 |c|^2 \cos^2 \theta = 1$$

$$\Rightarrow 3 = 4 \cos^2 \theta$$

$$\Rightarrow \cos^2 \theta = \frac{3}{4} \Rightarrow \cos \theta = \pm \frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{6}$$

102. (d) We know

$$|a + b + c|^2 = |a|^2 + |b|^2 + |c|^2 + 2(a \cdot b + b \cdot c + c \cdot a)$$

$$0 = 4 + 9 + 25 + 2(a \cdot b + b \cdot c + c \cdot a)$$

$$\Rightarrow \frac{-(4+9+25)}{2} = a \cdot b + b \cdot c + c \cdot a$$

$$\Rightarrow a \cdot b + b \cdot c + c \cdot a = -\frac{38}{2} = -19$$

103. (c) $a + b = 4\hat{i} + \hat{j} - \hat{k}$

$$a - b = -2\hat{i} + 3\hat{j} - 5\hat{k}$$

$$(a + b) \cdot (a - b) = |a + b||a - b|\cos\theta$$

$$-8 + 3 + 5 = \sqrt{16+1+1}\sqrt{4+9+25}\cos\theta$$

$$\Rightarrow 0 = \cos\theta \Rightarrow \theta = \frac{\pi}{2}$$

104. (b) We have

$$S = \{2, (1,4)\}$$

$$n(S) = 2$$

So Number of elements in power set P(S)

$$= n(P(S)) = 2^2 = 4$$

105. (a) $(1 - x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$

Put $x = 1$

$$(1 - 1 + 1)^n = a_0 + a_1 + a_2 + a_3 + \dots + a_{2n}$$

$$\Rightarrow 1 = a_0 + a_1 + a_2 + a_3 + \dots + a_{2n}$$

Put $x = -1$

$$[1 - (-1) + (-1)^2]^n = a_0 - a_1 + a_2 - a_3 + a_4 - \dots + a_{2n}$$

$$\Rightarrow 3^n = a_0 - a_1 + a_2 - a_3 + a_4 - \dots + a_{2n}$$

Add Eqs. (i) and (ii),

$$3^n + 1 = 2(a_0 + a_2 + a_4 + \dots + a_{2n})$$

$$\Rightarrow \frac{3^n + 1}{2} = a_0 + a_2 + a_4 + \dots + a_{2n}$$

106. (a) According to the questions

Required number of ways = Number of ways of putting p

Small animals in n cages x Number of ways of putting

Remaining animals in the cages

$$= {}^{m-n}P_p \times {}^{m-p}P_{m-p}$$

107. (c) given

$$n(A) = 4 \text{ and } n(B) = 2$$

$$\text{So } n(A \times B) = n(A) \cdot n(B) = 4 \times 2 = 8$$

$$\text{So Number of subsets of } A \times B = 2^{n(A \times B)}$$

$$= 2^8 = 256$$

So the number of subsets of the set $A \times B$, having atleast three elements is

$$= 256 - ({}^8C_0 + {}^8C_1 + {}^8C_2)$$

$$= 256 - (1 + 8 + 28) = 256 - 37 = 219$$

108. (b) $f(x) = x^2 \sin \frac{1}{x}, f(0) = 0$

$$\text{LHD} = \lim_{x \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \frac{(-h)^2 \sin(\frac{1}{-h})}{(-h)} = \lim_{h \rightarrow 0} \frac{\sin(\frac{1}{h})}{(\frac{1}{h})}$$

$$= \frac{-1 \leq \sin(\infty) \leq 1}{\infty} = 0$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

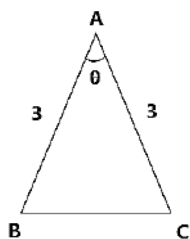
$$= \frac{h^2 \sin(\frac{1}{h})}{h} = \frac{\sin(\frac{1}{h})}{(\frac{1}{h})}$$

$$= \frac{-1 \leq \sin(\infty) \leq 1}{\infty} = 0$$

$$\text{LHD} = \text{RHD} = f'(0) = 0 = \text{Slope of function.}$$



109. (d)



Area of triangle ABC = $\frac{1}{2}(AB)(AC) \sin A$

$$\frac{1}{2} \times 3 \times 3 \sin \theta = \frac{9}{2} \sin \theta$$

Area will be maximum when, $\theta = \frac{\pi}{2}$

So maximum area of triangle ABC

$$= \frac{9}{2} \sin \frac{\pi}{2} = \frac{9}{2} \times 1 = \frac{9}{2}$$

110. (a) $\int_0^{\pi} x^3 \sin x dx$

Using integration by parts

$$= x^3[-\cos x] - \int (-\cos x) \cdot 3x^2 dx$$

$$= -x^3 \cos x + 3 \int x^2 \cos x dx$$

$$= -x^3 \cos x + 3[x^2(\sin x) - \int \sin x \cdot (2x) dx]$$

$$= -x^3 \cos x + 3x^2 \sin x - 6 \int x \sin x dx$$

$$= -x^3 \cos x + 3x^2 \sin x - 6[x(-\cos x) - \int (-\cos x) \cdot 1 dx]$$

$$= -x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x$$

$$\therefore \int_0^{\pi} x^3 \sin x dx = [-x^3 \cos x$$

$$+ 3x^2 \sin x + 6x \cos x - 6 \sin x]_0^{\pi}$$

$$= -\pi^3 \cos \pi + 3\pi^2 \sin \pi + 6\pi \cos \pi - 6 \sin \pi = \pi^3 - 6\pi$$

111. (a) Let $f(x)$ be a polynomial of degree four

$$f(x) = ax^4 + bx^3 + cx^2 + dx + e$$

$$\lim_{x \rightarrow 0} \left[1 + \frac{(ax^4 + bx^3 + cx^2 + dx + e)}{x^2} \right] = 3$$

$$\lim_{x \rightarrow 0} [1 + ax^2 + bx + c + d/x + e/x^2] = 3$$

Answer of limit is finite which is 3, hence the value of d and e must be zero.

Hence, limit is $(1 + 0 + 0 + c) = 3$

Hence, $c = 2$

So, $f(x)$ will be $ax^4 + bx^3 + 2x^2$.

$f(x)$ has extreme values at $x = 1$ and $x = 2$

$$f'(x) = 4ax^3 + 3bx^2 + 4x$$

$$f'(1) = 4a + 3b + 4 = 0$$

$$f'(2) = 32a + 12b + 8 = 0$$

Solving these equations,

$$a = 1/2, b = -2$$

So,

$$f(x) = (x^4/2) - 2x^3 + 2x^2$$

$$\text{So } f(2) = 0$$

112. (d)

$$4\sin^2 x + 3\cos^2 x + \sin\left(\frac{x}{2}\right) + \cos\left(\frac{x}{2}\right)$$

$$\Rightarrow 3 + \sin^2 x + \sin\frac{x}{2} + \cos\frac{x}{2}$$

$$\Rightarrow \text{max. value of } \sin\frac{x}{2} + \cos\frac{x}{2} = \sqrt{2}$$

$$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2} \Rightarrow x = 90^\circ$$

$$\Rightarrow 3 + \sin^2 90^\circ + \sin 45^\circ + \cos 45^\circ = 4 + \sqrt{2}$$

113. (d) We have,

$$(e^x + 1)y dy = (y + 1)e^x dx \Rightarrow \frac{y}{y+1} dy = \frac{e^x}{e^x+1} dx$$

On integrating both the sides, we get

$$\int \frac{y}{y+1} dy = \int \frac{e^x}{e^x+1} dx \Rightarrow \left(1 - \frac{1}{1+y}\right) dy = \int \frac{e^x}{e^x+1} dx$$

$$\Rightarrow y - \log(y+1) = \log(e^x + 1) + \log C$$

$$\Rightarrow y = \log(y+1) + \log(e^x + 1) + \log C$$

$$\Rightarrow y = \log[C(y+1)(e^x + 1)]$$

$$\Rightarrow e^y = C(y+1)(e^x + 1)$$

$$114. (b) \int_0^1 x(1-x)^n dx = \int_0^1 (1-x)(1-(1-x))^n dx$$

$$= \int_0^1 (1-x)x^n dx$$

$$= \left[\frac{x^{n+1}}{n+1} \right]_0^1 - \left[\frac{x^{n+2}}{n+2} \right]_0^1 = \frac{1}{n+1} - \frac{1}{n+2}$$

$$= \frac{n+2-n-1}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)}$$

115. (d) Critical point and nature for function

$$f(x, y) = x^2 - 2x + 2y^2 + 4y - 2$$

This is extrema for two variables, hence we use partial differentiation,

$$\text{Partial derivatives } \frac{\partial f}{\partial x} = 2x - 2 = 0$$

$$\Rightarrow x = 1 \text{ and } \frac{\partial f}{\partial y} = 4y + 4 = 0 \Rightarrow y = -1$$

So, critical points are $(1, -1)$

$$f''(x) = 2 > 0 \rightarrow \text{minima}$$

$$f''(y) = 4 > 0 \rightarrow \text{minima}$$

So, $f(x, y)$ has minimum at $(1, -1)$

116. (b)

$$\text{Given } y = (\cos x)^2$$

Using chain rule,

$$\frac{dy}{dx} = 2(\cos x)^{2-1} \frac{d}{dx} \cos x$$

$$= 2(\cos x)(-\sin x) \frac{dy}{dx} x^2$$

$$= 2(\cos x)(-\sin x) - 2x$$

(b) option match

117. (a) Let given function is

$$y = x^3 + e^x + 3^x + \cot x$$

Now

$$\frac{dy}{dx} = 3x^2 + e^x + 3^x \log 3 - \text{cosec}^2 x$$

118. (b) Given differential equation is

$$\frac{dy}{dx} = e^{x+y} + x^2 e^y = e^y (x^2 + e^x)$$

$$\Rightarrow \frac{dy}{e^y} = (x^2 + e^x) dx \quad [\text{Using variable separable form}]$$

Integrating both sides, we get

$$\int e^{-y} dy = \int (x^2 + e^x) dx \Rightarrow -e^{-y} = \frac{x^3}{3} + e^x + C$$

$$\Rightarrow e^x + e^{-y} + \frac{x^3}{3} = C \quad [\text{where, } C \text{ is constant}]$$

119. (a)

$$y = [-\log_e(\log_e x), x > 1]$$

On differentiation,

$$\frac{dy}{dx} = -\frac{d}{dx}(\log(\log x)) = -\frac{1}{\log x} \frac{d}{dx}(\log x) = -\frac{1}{x \log x}$$

$$120. (d) \lim_{x \rightarrow 0} \frac{x \tan x}{(1 - \cos x)} = \lim_{x \rightarrow 0} \frac{x \tan x}{1 - 1 + 2 \sin^2 \frac{x}{2}}$$

$$= \lim_{x \rightarrow 0} \frac{x \tan x}{\frac{2 \sin^2(\frac{x}{2})}{\frac{x^2}{4}}}$$

$$\lim_{x \rightarrow 0} \frac{x \tan x}{\frac{x^2}{2}} = \lim_{x \rightarrow 0} \frac{2 \tan x}{x} = 2$$



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